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| 5 | A graph of the voltage input to an ideal transformer is shown in Fig. 5.1. The 45:1 step-down transformer has a mean input power of 25 W. | | | |

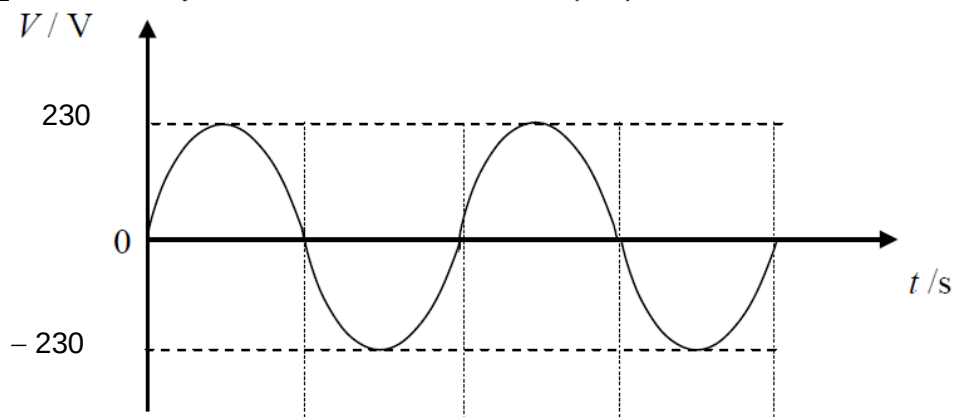
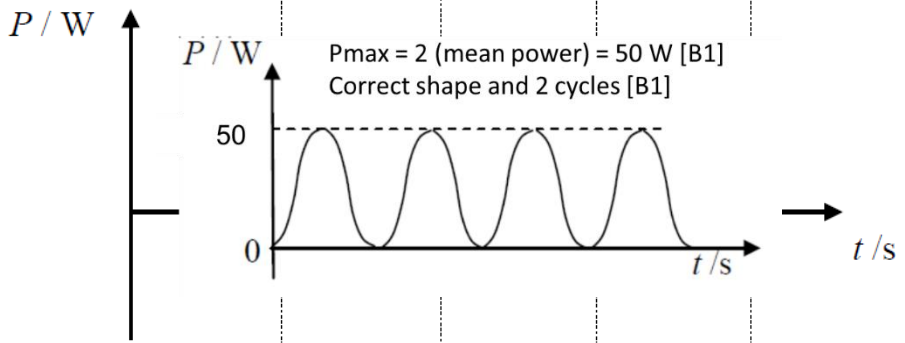


Fig. 5.1

			
			Fig. 5.2
	(a)	On the P - t graph in Fig 5.2, sketch the variation of power input to the transformer with time for two complete cycles. Indicate the value for the maximum input power.	[2]
	(b)	Calculate the r.m.s. value of the output voltage.	
		$\frac{V_{0,s}}{V_{0,p}} = \frac{N_s}{N_p}$ $V_{0,s} = V_{0,p} \frac{N_s}{N_p} = 230 \left(\frac{1}{45} \right) = 5.11 \text{ V}$ $V_{rms,s} = \frac{V_{0,s}}{\sqrt{2}}$ $= 3.6(1) \text{ V}$	
		of output voltage = V	[3]